

Polynomials, factor and remainder theorems - Set 1

TMUA practice questions by topic. Specification MM1.6. 12 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

1 When $x^3 - 4x^2 + kx + 6$ is divided by $(x - 3)$ the remainder is -6 . Find k .

- A 1
- B 5
- C -3
- D -5
- E 3
- F -1

2 $(x + 2)$ is a factor of $2x^3 + ax^2 - 5x - 6$. Find the value of a .

- A 5
- B -2
- C 1
- D -3
- E 2
- F 3

3 Fully factorise $x^3 - 7x + 6$.

- A $(x - 1)(x - 2)(x - 3)$
- B $(x - 1)(x - 2)(x + 3)$
- C $(x - 1)(x + 2)(x - 3)$
- D $(x - 1)(x^2 + x + 6)$
- E $(x + 1)(x + 2)(x - 3)$
- F $(x + 1)(x - 2)(x + 3)$

- 4 Find the remainder when $x^4 + 2x^3 - 3x + 1$ is divided by $(2x - 1)$.
- A $\frac{13}{16}$
 B $\frac{16}{16}$
 C 1
 D $-\frac{1}{4}$
 E $-\frac{16}{16}$
 F $-\frac{1}{16}$
- 5 The polynomial $p(x) = x^3 + ax^2 + bx + 4$ has $(x - 1)$ and $(x + 2)$ as factors. Find the third linear factor.
- A $(x + 1)$
 B $(x - 2)$
 C $(x - 4)$
 D $(x - 1)$
 E $(x + 4)$
 F $(x + 2)$
- 6 Find the quotient when $2x^3 - 3x^2 + 4x - 5$ is divided by $(x - 1)$.
- A $2x^2 + x - 3$
 B $2x^2 - 5x + 9$
 C $2x^2 - x + 3$
 D $2x^2 - x - 3$
 E $2x^2 + x + 3$
 F $2x^2 - x + 5$
- 7 Find the coefficient of x^2 in the expansion of $(x + 1)(x - 2)(2x + 3)$.
- A 1
 B -3
 C -2
 D -1
 E 2
 F 3

- 8 Which of the following is a factor of $x^4 - 5x^2 + 4$?
- A $x - 3$
 - B $x^2 + 1$
 - C $x - 4$
 - D $x + 4$
 - E $x + 2$
 - F $x^2 + 4$
- 9 $p(x)$ is a cubic polynomial with $p(1) = p(2) = p(3) = 0$ and $p(0) = 12$. Find $p(4)$.
- A 24
 - B -24
 - C 12
 - D 0
 - E -12
 - F 6
- 10 The polynomial $x^3 + ax^2 + bx + c$ has roots 2, -1 and 3. Find the value of $a + b + c$.
- A 5
 - B -5
 - C -3
 - D 3
 - E -7
 - F 7
- 11 $p(x)$ is a polynomial. P : ' $p(2) = 0$ '. Q : ' $(x - 2)^2$ is a factor of $p(x)$ '. Which one of the following is true?
- A P is necessary and sufficient for Q .
 - B P is necessary but not sufficient for Q .
 - C P is sufficient but not necessary for Q .
 - D P is neither necessary nor sufficient for Q .

- 12** When $p(x)$ is divided by $(x - 1)$ the remainder is 3, and when divided by $(x + 1)$ the remainder is -1 . Find the remainder when $p(x)$ is divided by $(x^2 - 1)$.
- A** 2
- B** $2x + 1$
- C** $-x + 2$
- D** $2x - 1$
- E** $x + 2$
- F** $3x - 1$